

V SHRAVANTH

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SKILLS

Languages: Python, JavaScript, SQL, Rust

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Agentic AI: OpenAI, Anthropic, Gemini, OpenRouter, Groq; Tool calling, RAG, fallback routing, context injection, LLM evaluation

Backend: FastAPI, Pydantic, Uvicorn, Async HTTPX, Rate Limiting, JWT Auth

Frontend: Chrome Extension (Manifest V3), Vanilla JS, DOM Manipulation

Infra/Tools: Vercel (Serverless), Vercel KV (Redis), Git, Docker, Supabase, Linux

PROJECTS

ClearPath — RAG-Powered AI Support System

Live | GitHub

- **Tools:** Python, FastAPI, BM25Okapi, Llama 3.1 8B / Llama 3.3 70B.
- Built a production-deployed RAG support chatbot on a three-layer backend (retrieval, routing, evaluation), each layer addressing a distinct failure mode most implementations ignore.
- Implemented BM25Okapi scoring with word-level chunking over a 30-document PDF corpus for context-aware retrieval that surfaces the correct document, not just the nearest semantic match.
- Engineered a deterministic router that sends simple queries to Llama 3.1 8B and complex ones to 3.3 70B, plus an output evaluator that flags missing context, refusals, and internal-data leaks before responses reach the user.

Spark — Personal AI Agent (Telegram)

GitHub | LinkedIn Post

- **Tools:** Python, FastAPI, Telegram Bot API, OpenRouter, Vercel Serverless, Vercel KV, GitHub REST API, Gmail API, OAuth2.
- Built a fully deployed personal AI agent with 43 tools across GitHub (26), Gmail (17), web search, and notes in one Telegram thread, with keyword routing that exposes only relevant tools per request instead of all 43.
- Engineered a full Gmail OAuth2 flow with token persistence via Vercel KV (Redis) and a multi-round agent loop supporting up to 5 chained tool calls per request.
- Deployed on Vercel serverless with automatic LLM fallback across Llama 3.3 70B, Mistral Small 3.1, and Nvidia Nemotron via OpenRouter — zero server maintenance, runs 24/7 for free.

SubSync — Context-Aware AI Orchestration System

Live | Webstore

- **Tools:** FastAPI, Python, JavaScript, OpenRouter API, Groq, MegaLLM, Pydantic, HTTPX, Vercel, Supabase, LocalStorage.
- Engineered Hierarchical Local Context Compression (HLCC), achieving 90–97% token reduction for massive conversations via client-side intent extraction.
- Built a high-availability 3-tier fallback architecture (Primary → Fallback 1 → Fallback 2) with automated provider failover and thread-safe token tracking.
- Designed a Prompt Improver using DOM mutation observers to inject real-time prompt engineering into native chat UIs; Supabase/Postgres waitlist with Row-Level Security.

AirWrite — AI Voice-to-Text Desktop Assistant

GitHub

- **Tools:** Rust, Tauri 2, React 18, Vite, Tailwind CSS 4, Groq Whisper, CPAL, Windows Credential Manager.
- Built a Windows desktop dictation assistant with global hotkey support that records speech from any application and pastes transcriptions into the focused window via simulated **Ctrl+V**.
- Engineered a low-latency speech pipeline using Groq Whisper Turbo with optional Llama 3.3 70B post-processing for grammar correction, punctuation cleanup, and filler-word removal.
- Implemented production-grade reliability — audio ducking with crash recovery, clipboard restoration, silence detection, push-to-talk/toggle modes, waveform overlay, and dictation history — entirely in Rust + Tauri.

EDUCATION

Scaler School of Technology

2023 – 2027

Integrated CS Program (Bachelors + Masters)

BITS (Birla Institute of Technology and Science)

2023 – 2026

B.S. Computer Science